Dilutions of Stock Solutions Worksheet -Answers

1. Use the dilution equation: $M_c \bullet V_c = M_d \bullet V_d$

$$V_c = \frac{M_d \bullet V_d}{M_c} = \frac{0.2\ M \bullet 2.00\ L}{17.8\ M} = 0.022\ L = 22\ mL$$

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2. a) Use the dilution equation: $M_c \bullet V_c = M_d \bullet V_d$

$$M_d = \frac{M_c \bullet V_c}{V_d} = \frac{0.05\ M \bullet 5\ mL}{100\ mL} = 0.0025\ M$$

b) $\frac{0.0025\ M}{1000\ mL} = \frac{x}{10\ mL}$ $x = 0.000025$ mol of $CuSO_4$

$m = n \bullet M = 0.000025$ mol $\bullet$ 159.62 g/mol = 0.00399 grams = 4 mg

c) The amount required is too small to measure on a laboratory scale. Microgram scales are available but the cost is prohibitively high.

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3. a) The concentration of the dilute solution is only 4% that of the original solution.

b) The volume has increased by a factor of 25 times.

c) The speed should be 25 times slower.

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4. $M = \frac{0.274\ g}{L}$

$$M_c = \frac{M_d \bullet V_d}{V_c} = \frac{0.274\ g/L \bullet 250\ mL}{10\ mL} = \frac{6.85\ g}{L}$$

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5. a) $\frac{0.100\ mol}{1000\ mL} = \frac{x}{100\ mL}$ $x = 0.01$ moles needed

$m = n \bullet M = 0.01$ mol $\bullet$ 237.95 g/mol = 2.38 grams of $CoCl_2 \bullet 6H_2O$

b) $M_c \bullet V_c = M_d \bullet V_d$

$$V_c = \frac{M_d \bullet V_d}{M_c} = \frac{0.01\text{ M} \times 100\text{ mL}}{0.1\text{ M}} = 10\text{ mL of concentrate needed}$$

c) To a clean 100 mL volumetric flask add 50 mL of distilled H_2O. To this add 2.35 grams of cobalt(II) chloride hexahydrate. Swirl to dissolve. Top up to 100 mL mark with distilled water. This the stock solution at 0.1 M. Using a 10 mL volumetric pipette remove 10 mL of 0.1 M solution and place it in another clean 100 mL columetric flask. Add distilled water to this second flask until it reaches the 100 mL volumetric flask mark. This is the second solution.

Apparatus - 2 100 mL volumetric flask, 2.38 g of $CoCl_2 \bullet 6H_2O$
Safety - DO NOT PIPETTE BY MOUTH

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6. **a)** $M_c \bullet V_c = M_d \bullet V_d$

$$V_c = \frac{M_d \bullet V_d}{M_c} = \frac{0.01\text{ M} \times 100\text{ mL}}{0.4\text{ M}} = 25.0\text{ mL of concentrate needed}$$

Obtain the bottle of 0.4 M stock solution. Pour some into a clear dry beaker. (Never pipette directly from the stock bottle) Using a 25.0 mL volumetric flask, pipette 25.0 mL of stock solution from the beaker. Transfer this to a clear dry 100 mL volumetric flask. Add water to this flask until the water reaches the 100 mL volumetric mark.
Apparatus - 1 - 25 mL volumetric pipette
1 - 100 mL volumetric flask
1 - stock solution
1 - beaker (50 mL size recommended)

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7. $M_c \bullet V_c = M_d \bullet V_d \quad V_d = \frac{M_c \bullet V_c}{M_d} = \frac{6.0\text{ M} \bullet 30.0\text{ mL}}{150.0\text{ mL}} = 1.2\text{ M}$

answer a)

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8. $M_c \bullet V_c = M_d \bullet V_d \quad V_d = \frac{M_c \bullet V_c}{M_d} = \frac{8.0\text{ M} \bullet 40.0\text{ mL}}{160.0\text{ mL}} = 2.0\text{ M}$

answer d)

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9. $M_c \bullet V_c = M_d \bullet V_d \quad (V + 555\ mL) = \frac{18.0\ M \bullet V\ mL}{2.4\ M\ mL}$

$V = 85\ mL$

answer b)

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10. $M_c \bullet V_c = M_d \bullet V_d \quad M_d = \frac{M_c \bullet V_c}{V_d} = \frac{6.0\ M \bullet 75.0\ mL}{150\ mL} = 3\ M$

answer c)

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11. $V_c = \frac{M_d \bullet V_d}{M_c} = \frac{0.15\ M \bullet 4.2\ L}{6.0\ M} = 0.105\ L = 105\ mL$

answer c)

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12. $M_c \bullet V_c = M_d \bullet V_d \quad (V + 800\ L) = \frac{0.130\ M \bullet 800\ L}{0.100\ M}$

$V = 240\ L$

answer d)